Smart Room-Light System Using Sensors

Arduino Uno • LDR • PIR Motion Sensor • LED • Buzzer

Project status: Documentation and starter code prepared from the supplied circuit screenshot. Actual simulation tests still need to be run and recorded; no test results are fabricated.

1. Objective

To build an automatic room-light controller that reads ambient light using an LDR and detects movement using a PIR sensor. The room-light LED turns on only when motion is detected and the room is dark. A buzzer provides a motion alert. The screenshot also contains a TMP36 temperature sensor, retained as an optional extra alert.

2. Components

Component	Purpose
Arduino Uno	Reads sensors and controls outputs
LDR + fixed resistor	Ambient light measurement through a voltage divider
PIR sensor	Motion detection
LED + 220–330 Ω resistor	Represents the room light
Buzzer	Audible motion alert
TMP36 (optional)	Additional temperature reading and alert
Breadboard/jumper wires	Circuit assembly

3. Intended pin mapping

Component	Arduino connection
LDR divider output	A0
PIR OUT	D2; VCC to 5V; GND to GND
LED	D9 through 220–330 Ω resistor
Buzzer	D8 and GND
TMP36 output (optional)	A1; connect power pins according to its datasheet/simulator pinout

Note: Screenshot wire junctions are not all unambiguous. This mapping is the intended mapping for the included source code and must be checked in Tinkercad.

4. System logic

The Arduino reads the LDR on A0 and PIR output on D2. With the default setting, an LDR reading above 500 is classified as dark. The LED turns ON only when PIR motion is HIGH and the dark condition is true. The buzzer turns ON when motion is detected or the optional TMP36 temperature is above 30 °C. The LDR threshold and reading direction must be calibrated using observed bright and dark readings.

5. Test plan and expected results

Ambient light	Motion	Expected LED	Expected buzzer
Bright	No	OFF	OFF*
Dark	No	OFF	OFF*
Dark	Yes	ON	ON

Ambient light	Motion	Expected LED	Expected buzzer
Bright	Yes	OFF	ON

*Buzzer may also sound if TMP36 temperature exceeds 30 °C. These are expected outcomes from the code, not verified results. Enter actual readings and pass/fail status in the CSV and test report after simulation.

6. Observations, limitations and improvements

• Observation fields to complete: bright LDR reading, dark LDR reading, chosen threshold, PIR response time, and any unexpected output behaviour.

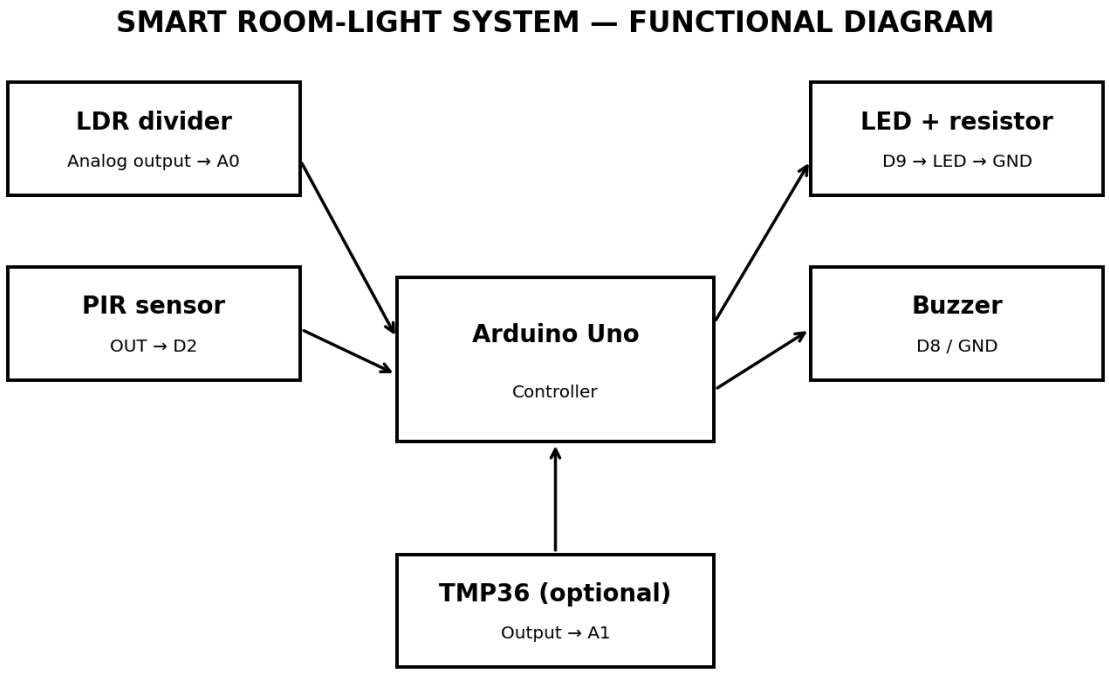
• Limitations: LDR readings vary with lighting and resistor values; PIR sensors need warm-up and may have response delay; this prototype uses an LED rather than a real room lamp.

• Possible improvements: add a configurable time-out so the light remains on briefly after motion; add an LCD/OLED; log readings; use hysteresis to prevent flickering near the threshold; use a properly rated isolated driver for a higher-power lamp.

7. Safety

Use a current-limiting resistor with the LED. Never connect mains-powered lighting directly to an Arduino pin. A real lamp requires properly rated, isolated switching hardware and safe supervised installation.

8. Circuit diagram



All grounds common; use current-limiting resistor with LED.

9. Evidence checklist

Included: supplied circuit screenshot. Still required: Serial Monitor screenshots and actual test observations for bright/dark and motion/no-motion conditions. The included circuit screenshot alone does not prove successful testing.